Gemily Wang

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EDUCATION

University of California, San Diego — **September 2025 - Present**

Master of Science in Bioengineering

Relevant Coursework: Mathematical Methods for Bioengineering, Biochemistry, Cellular and Molecular Biology, Bioinformatics Algorithms, Systems Biology: Large-Scale Data Analysis

Northwestern University — **September 2020 - June 2024**

Bachelor of Science in Biomedical Engineering, Minors in Data Science and Global Health

Relevant Coursework: Data Science Sequence, Biostatistics, Biomedical Applications in Machine Learning, Computational Genomics

SKILLS

- **Computational:** MATLAB, Python, R, Microsoft Office, GraphPad Prism, CAD, Snapgene, FlowJo
- **Languages:** Chinese (fluent), Spanish (fluent), Korean (basic)

EXPERIENCE

Cell Line Development Intern — *Genentech*

June 2024 – June 2025 — *South San Francisco, CA*

- Developed and evaluated plasmids using Golden Gate cloning, performed full cycle from plasmid construction and cell transfection to production and analysis, comparing titer, product quality, and other quality metrics with traditional 6-way ligation methods
- Utilized CRISPRa technology to activate genes aimed at increasing titer and glycosylation in combination, optimizing antibody yields and glycosylation profiles
- Applied CRISPR-KI to modify a cell line's genome to prevent mutation-driven amino acid changes and enhance genetic stability
- Leveraged stress molecules to gradually acclimate cells to stressful conditions, evaluating their impact on titer

Teaching Assistant (Data Science) — *Department of Statistics and Data Science, Northwestern University*

September 2023 – June 2024 — *Evanston, IL*

- Provided academic support by assisting over 200 students in understanding course content and completing assignments during classes and weekly office hours
- Mentored five project teams by providing feedback on group project proposals, presentations, and reports
- Conducted weekly hour-long review sessions covering course theory and Python, emphasizing key concepts
- Demonstrated strong communication and organizational skills

Biomedical Engineering Capstone Design Project — *Department of Biomedical Engineering, Northwestern University*

September 2023 – March 2024 — *Evanston, IL*

- Collaborated in a group of four to design, construct, and test an MRI-safe deformable abdominal phantom for end-to-end testing of the MRI-Linac system
- Aim was to accurately assess deformations in guided radiation therapy through the calibration of the MRI machine and lead to improved cancer treatment outcomes
- Phantom design was a rectangular prism acrylic casing with a silicone stomach, a silicone duodenum, and a 3D printed pancreas submerged in mineral oil to create contrast and enhance image quality
- Stomach and duodenum were attached with silicone tubes long enough to reach outside the zone IV room so that the phantom operator could fill the organs with air or water to induce deformation
- Partnered with Dr. Poonam Yadav, assistant professor in radiation oncology at Northwestern Medicine

Immunophenotyping Intern *HiberCell (Biotech Company)*
June 2022 – August 2022 *Roseville, MN*

- Executed and analyzed Western Blot assays using Jess to monitor possible biomarkers on samples of different cell types including purified and unpurified MDSC (myeloid-derived suppressor cell) and PBMC (peripheral blood mononuclear cell)
- Explored methods on improving cell purification methods on tumor and organ tissues of multiple mouse models for analysis of pharmacodynamic markers in specific cell subsets through flow cytometry

June 2021 – August 2021

- Executed sandwich ELISA (enzyme-linked immunosorbent assay) with human plasma samples to detect and measure subject-stratifying biomarkers for monitoring immune-oncology therapy
- Executed sandwich ELISA with plasma of tumor-bearing mice post multiple cancer drug treatments to identify possible pharmacodynamic biomarkers related to drug efficacy
- Performed RNA isolation, first strand cDNA syntheses, and TaqMan qPCR assay to study specific gene expressions from different tissues of in vivo mouse models after immunotherapy cancer drug treatments
- Worked with the animal research and immunophenotyping teams to help with terminations and processing mice tissues for flow cytometry

PUBLICATIONS

Wan J, Patino S, Young P, **Wang G**, et al. Development of Multiplexed CRISPRa Activation Technique to Enhance Antibody Production in CHO Cells. (in submission)

Wan J, Young P, Patino S, **Wang G**, et al. Improve recombinant protein production in CHO cells by reading and writing m6A-mRNA methylation. (in submission)

POSTERS

Drees J, Chan AS, et al., **Wang G**, et al. Novel GCN2 Modulator HC-7366 Decreases Pulmonary Metastases and Reduces Myeloid-Derived Suppressor Cells. *Journal for ImmunoTherapy of Cancer*. 2022; 10. doi: 10.1136/jitc-2022-SITC2022.1327.

Maldonado C, **Wang G**, et al. A Novel Abdominal Phantom with Deformable Organs. Poster presented at: 2024 AAPM Annual Meeting; July 2024; Los Angeles, CA.